Problems with HTML, CSS, JavaScript (Before React)

When applications became large (like Facebook, Gmail), using only HTML, CSS, JS had problems:

1. Full Page Reload

- Every click → Entire page reload

2. DOM Manipulation is Difficult

In JavaScript:
- document.getElementById("title").innerHTML = "Hello";

Example:
- For big applications: Too many DOM operations. Hard to manage

3. No Code Reusability

In HTML:
- No Code Reusability.
- No proper component system

4. Poor Performance for Large Apps

- Updating DOM directly is slow.

Why React Came into Picture?
- To solve above problems.
- React was developed by Meta.
- It was first used inside Facebook application.

Goal:
- Faster UI updates
- Better performance
- Component-based structure
- Easy maintenance

React:

- A JavaScript library used to build fast and interactive user interfaces using components.

It mainly builds:
- Single Page Applications (SPA)
- Dynamic web apps

1. Component-Based Architecture
- UI is divided into small reusable parts.

Example:
- Header component
- Footer component
- Sidebar component

Reusable and clean structure.

2. Virtual DOM (Fast Performance)
- React uses Virtual DOM.
- It updates only changed parts.

Result:

Faster applications

3. Reusability

Create one component → Use multiple times.
Example:

```
<Button />
<Button />
<Button />
```

Saves time and reduces code duplication.

4. Single Page Application Support

No full page reload.

SPA is a web application that loads a single HTML file (usually index.html) once, and then dynamically updates content using JavaScript without reloading the entire page.

Component:

A Component is a small, reusable piece of UI (User Interface).

Component = Small block of UI
React application = Collection of components

A React component is a JavaScript function (or class) that returns JSX, which looks like HTML.

There are mainly 2 types:

1. **Functional Component**
2. **Class Component**

Functional Component:

Welcome.jsx

```
function Welcome() {
  return <h1>Welcome to react</h1>;
}
export default Welcome;
```

Use **Welcome Component** inside App:

```
function App() {
  return (
    <div>
      <Welcome />
    </div>
  );
}
```

<Welcome /> is a component.